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Undergraduate researcher, joint robot-hand team of the Sugano Laboratory and the Ogata Laboratory, Waseda University, Tokyo

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RESEARCH INTERESTS

Dexterous robot hands · hardware–controller co-design · teleoperation and retargeting · imitation learning

EDUCATION

Waseda University, Tokyo — B.Eng. in Modern Mechanical Engineering, School of Creative Science and Engineering — Apr 2023 – Mar 2027 (expected)

- Joint robot-hand team of the Sugano Laboratory (Prof. Shigeki Sugano) and the Ogata Laboratory (Prof. Tetsuya Ogata), Apr 2026 – present.
- Master's program in Modern Mechanical Engineering, Waseda University, from Apr 2027 (planned).

PUBLICATIONS

[1] **K. Sasazaki**, S. Funabashi, P. Ponnivalavan, A. Schmitz, T. Ogata, and S. Sugano, "PACT: Posture-Aligned Co-Design Technique for Hand Configuration and Retargeting in Multi-Finger Imitation Learning," arXiv:2610.18443, 2026. Under review. arxiv.org/abs/2610.18443

RESEARCH EXPERIENCE

Joint robot-hand team, Sugano and Ogata Laboratories, Waseda University — Apr 2026 – present

- **PACT Hand (first author).** Formulated and implemented the joint optimization of the mechanical parameters of a 17-servo anthropomorphic hand (link lengths, joint offsets, mounting angles, base placements) and its posture retargeter against human grasping postures from DexYCB, under servo-interference, assembly-clearance and flat-palm constraints. Designed the parts in Autodesk Fusion and built the hand (Dynamixel servos, 3D-printed parts).
- Built the teleoperation pipeline (single RGB camera → WiLoR hand pose → posture-IK retargeting) and evaluated the hand against the LEAP Hand: 3.9 mm vs 12.8 mm keypoint RMSE, 2 kg vs 1 kg payload under matched per-joint torque limits, 7× less deformation of a held paper cup, 86% vs 64% grasp success on YCB objects.
- Collected teleoperated demonstrations (GELLO leader arm + PACT Hand) and trained an ACT policy: 75% success at held-out placements in a can pick-and-place task.
- Teleoperation systems on ROS 2: LEAP Hand with a Manus glove; Allegro Hand with a SenseGlove Nova 2.
- Imitation-learning experiments: paper cutting with scissors on ALOHA; ACT, Diffusion Policy and SmolVLA compared on an SO-101 arm with LeRobot.

PROJECTS AND ACTIVITIES

ROBOSTEP, Waseda University robotics club (11th cohort) — mechanism design — Apr 2023 – present

- NHK Student Robocon 2025: mechanism design for the Waseda team, which reached the top 8.

Teaching assistant, Introduction to C Programming, Waseda University — Spring 2026

SKILLS

Programming: Python, C, ROS 2, Linux
Robot learning and simulation: LeRobot (ACT, Diffusion Policy, SmolVLA), MuJoCo
Hardware: Autodesk Fusion (CAD, API scripting), 3D printing, Dynamixel servos, mechanism design
Languages: Japanese (native), English (TOEIC L&R 870; EIKEN Grade Pre-1)

CERTIFICATIONS AND OTHER

- Professional Engineer (Japan) first-stage examination, mechanical engineering — passed, Feb 2026.
- Ski Association of Japan (SAJ) Grade 2; ski instructor at NASPA New Otani, winter seasons 2023–2025.

Updated October 2026